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Hardware Accelerators for AI

(based on the surveys of S M Mojahidul Ahsan, Anurag Dhungel, Mrityika Chowdhury, Md Sakib Hasan, Tamzidul Hoque, 2025)

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Bottleneck of the Traditional CPUs

- Von Neumann Architecture
 - Limited parallel processing capabilities
 - Narrower vector processing unit
 - Low memory bandwidth
 - Complex cache hierarchy introduce latency
 - General-purpose instruction set leads to underutilization in predictable, repetitive tasks
 - Inefficiencies in managing high volume of MAC (multiply-accumulate) operations
 - High power consumption

GPUs: A steppingstone but not enough

- Advantages
 - Excel at parallel processing with thousands of cores
 - Cutting-edge memory solutions: HBM, GDDR7
- Disadvantages
 - Due to the energy-intensive nature of AI algorithms, GPUs still expose limitation in power efficiency, memory bandwidth, and latency.

Specialized hardware accelerators for AI

- Key aspects
 - **High parallelism** for massive volume of MAC operation
 - **Near-memory processing, in-memory computing**
 - **High memory bandwidth** for rapid transfer of data between memory and processing units
 - **Energy efficiency**
 - **Reduce Instruction Set Computing (RISC)** for simpler, dedicate instructions for AI algorithms → faster execution and reduced power consumption
 - **Reduce precision arithmetic:** sacrificing accuracy for computational efficiency and memory bandwidth

Specialized hardware accelerators for AI

- Solutions

- GPUs

- Strengths: Popularity, high data throughput and massive parallelism, suitable for training AI models.
 - Weaknesses: Complexity and scalability

- **FPGAs (Field-programmable Gate Arrays)**

- Strengths: Reconfigurable hardware for specific AI tasks, enhanced performance and energy efficiency, well-suited for low-latency inference in real-time applications, beneficial for edge computing and IoT applications.
 - Weakness: programming complexity (VHDL, VeriLog), resource constraints, scalability

- **Application-specific Integrated Circuits (ASICs)**

- Strengths: exceptional energy efficiency, better performance
 - Weaknesses: lack of flexibility, high development cost

- **Neuromorphic Integrated Circuits:**

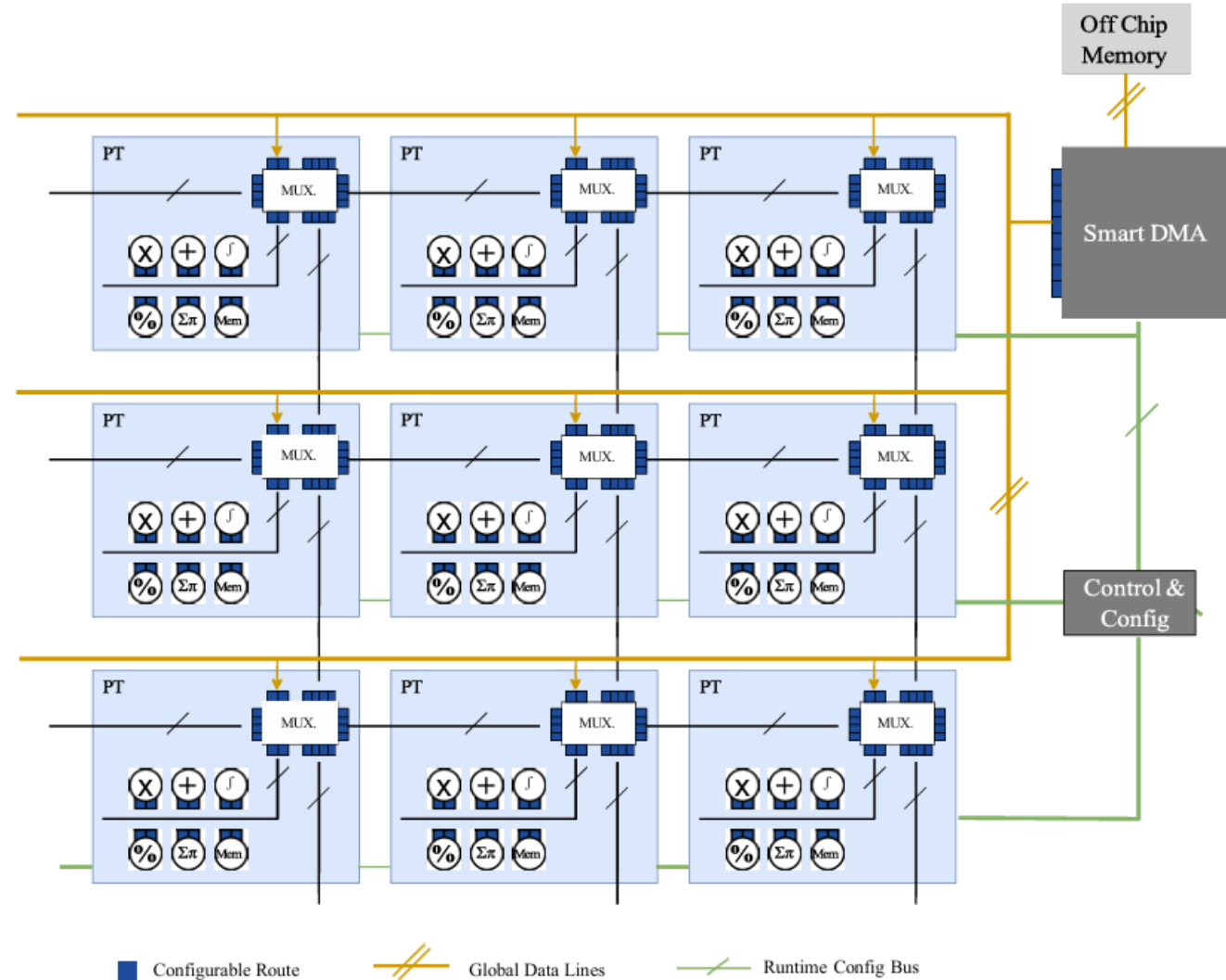
- Strengths: lower power consumption, faster data processing. Ideals for edge computing, IoT application
 - Weaknesses: Still in early development stages, challenge in programming and integration.

Specialized hardware accelerators for AI

- Architectures
 - NeuFlow architecture
 - The DianNao Series
 - The Neural Processing Unit (NPU)
 - RENO architecture
 - NeuroCube architecture
 - PRIME
 - Tensor Processing Unit (TPU)
 - Eyeriss v2
 - ComAct
 - ...

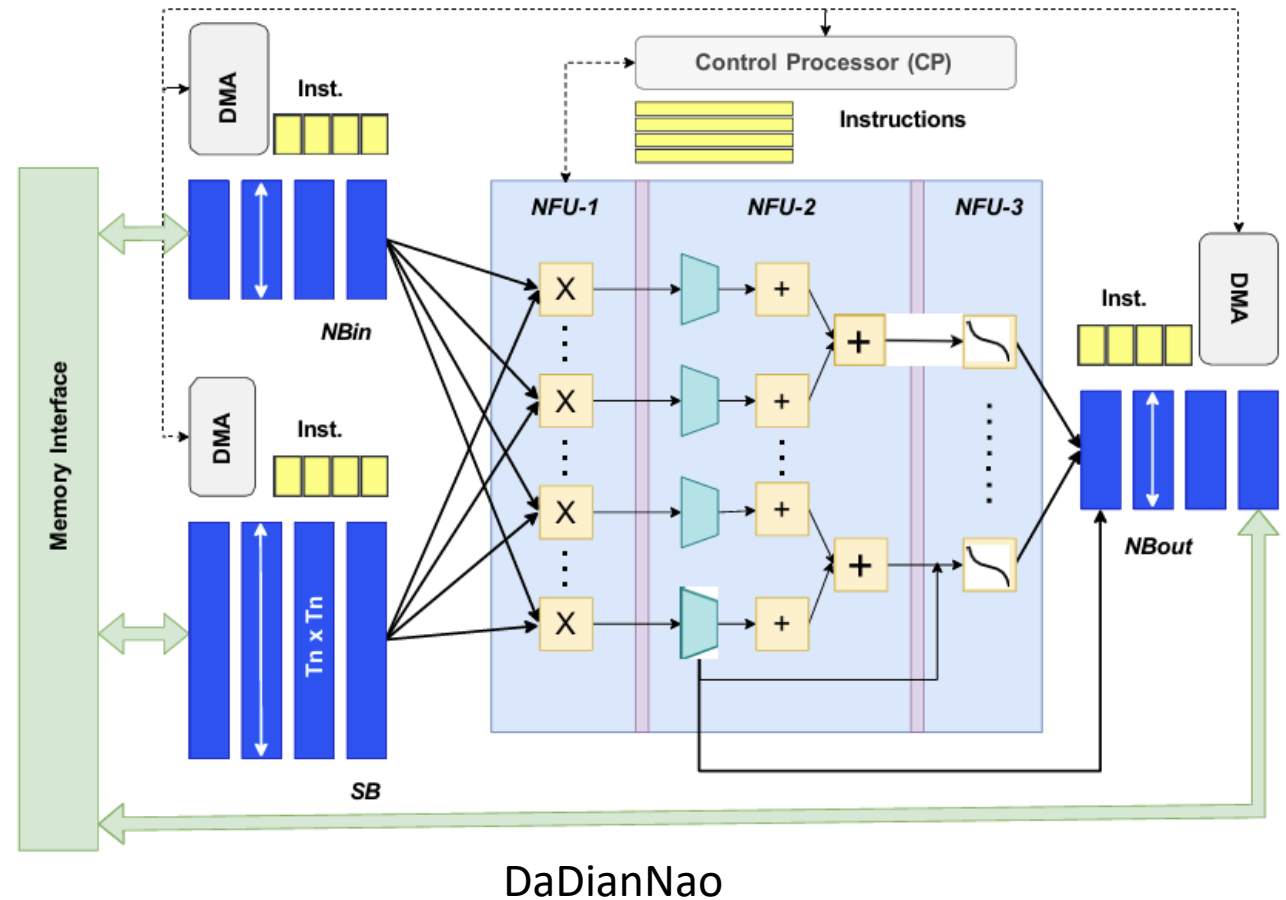
NeuFlow Architecture

- Introduced in 2011 for visual-related task, implemented in FPGA
 - High-level programming with LuaFlow compiler
 - Highly parallelism on 2D grid of PTs (Processing tiles)
 - Handles off-chip memory interaction with SmartDMA
 - Optimized for real-time vision tasks, high throughput for filter-based algorithms.



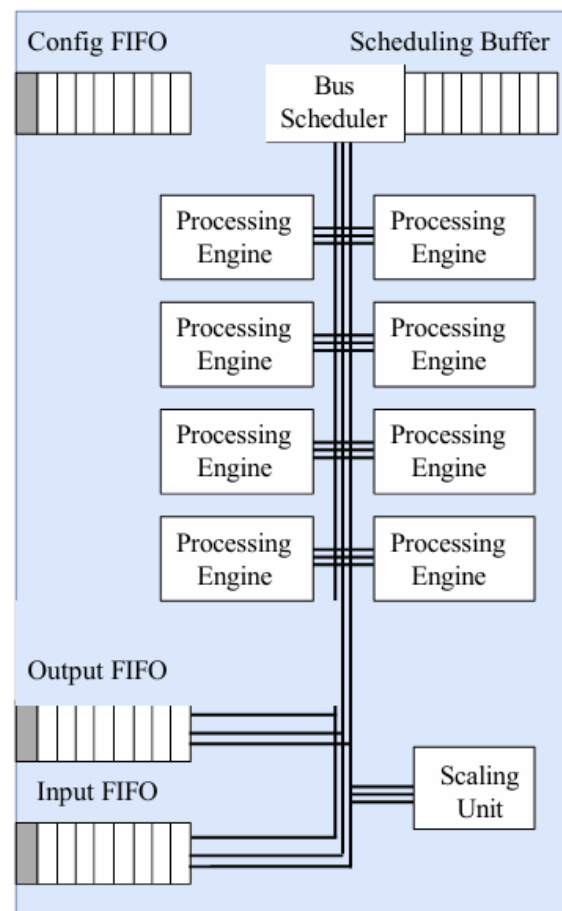
The DianNao Series

- DianNao, DaDianNao, ShiDianNao, PuDianNao
 - ASIC
 - Each architecture is specialized for a specific aspect of ML.
 - DianNao: Neural network
 - DaDianNao: Large-scale neural network
 - ShiDianNao: Low-powered, embedded system
 - PuDianNao: Extends to broader range of ML techniques

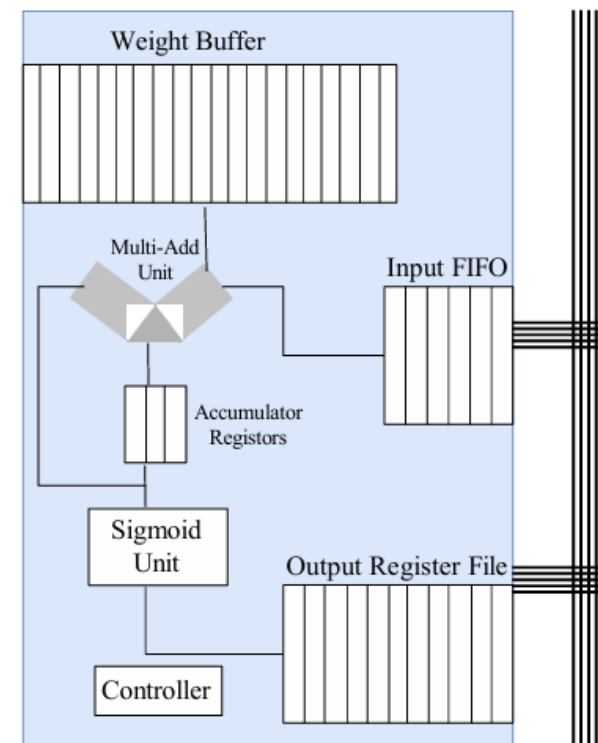


The Neural Processing Unit (NPU)

- Classified as ASIC but has integrated into traditional computing system as co-processor model
 - CPU identifies a code segments within a general-purpose program that are suitable for approximation
 - Parrot transformation: transforms the select segments into NN models.
 - Execution on NPU
 - Integrate results back to CPU
- Performance and Energy efficiency



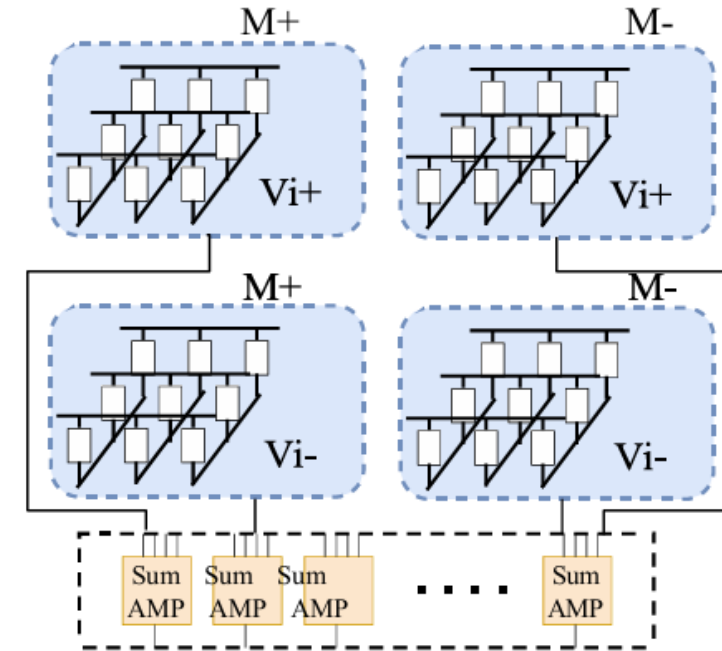
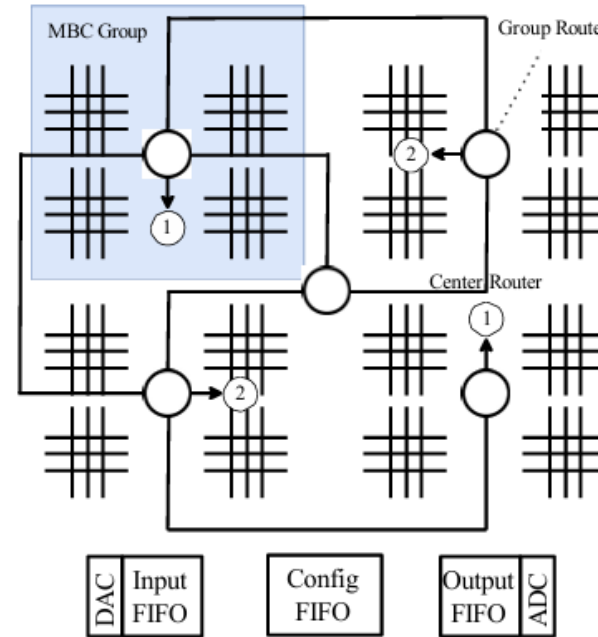
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(b)

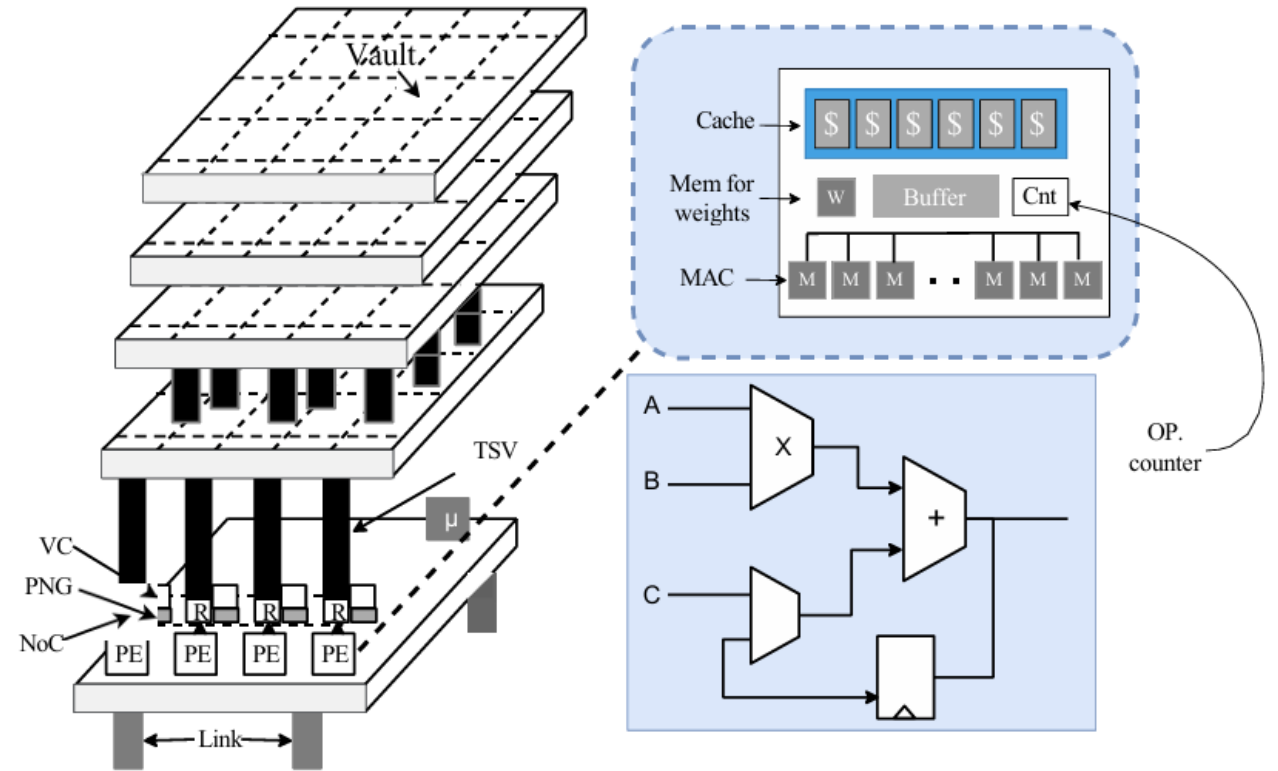
RENO architecture

- Classified as Neuromorphic IC
 - Spike neural network, reconfigurable
 - Memristor-based crossbar (MBC) arrays: efficient to memory storage and computation
 - Hierarchical architecture: flexible and efficient handling various ANN topologies
 - Mix-signal Interconnection Network (M-Net): facilitate communication between different layers.
- Performance and Energy efficiency
- Balancing Flexibility and Fault Tolerance

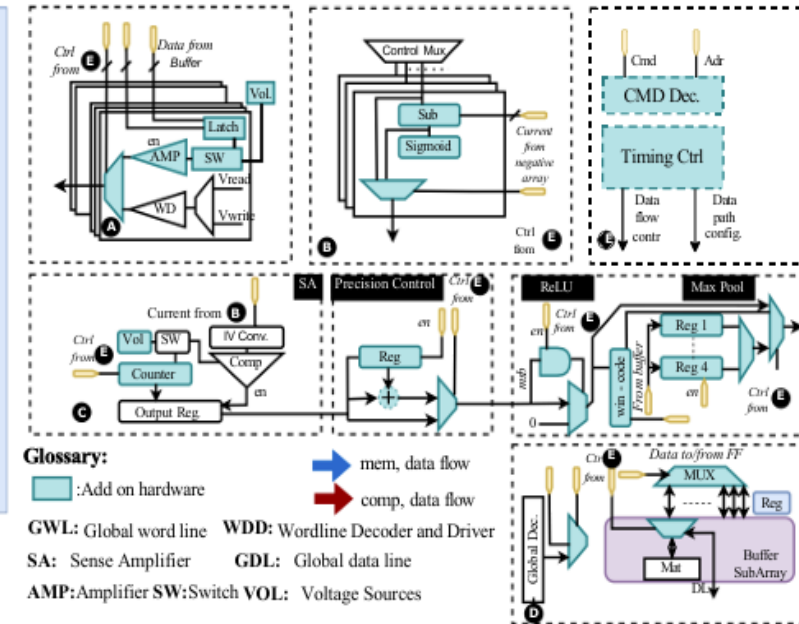
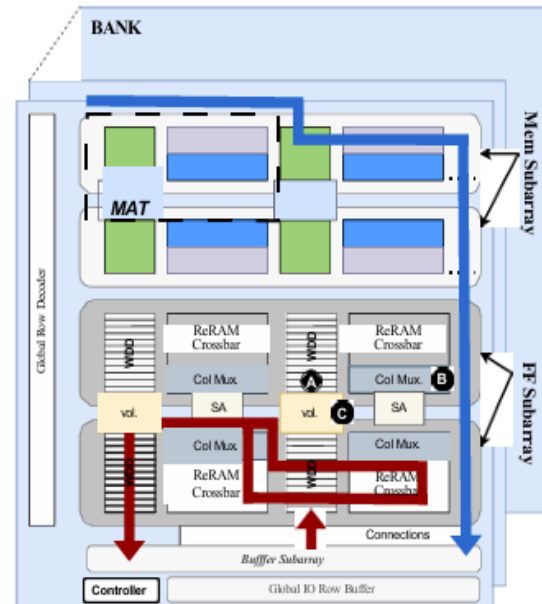


Neurocube architecture

- Classified as Neuromorphic IC
 - In-memory Neuromorphic Processing: combining computing layers within 3D memory stacks → reduce data movement
 - Memory-centric Neural Computing (MCNC): computation and energy efficiency
 - Programmable NeuroSequence Generator: flexibility
- Performance and Energy efficiency
- Boost memory bandwidth and minimize latency
- Challenges in design and manufacturing

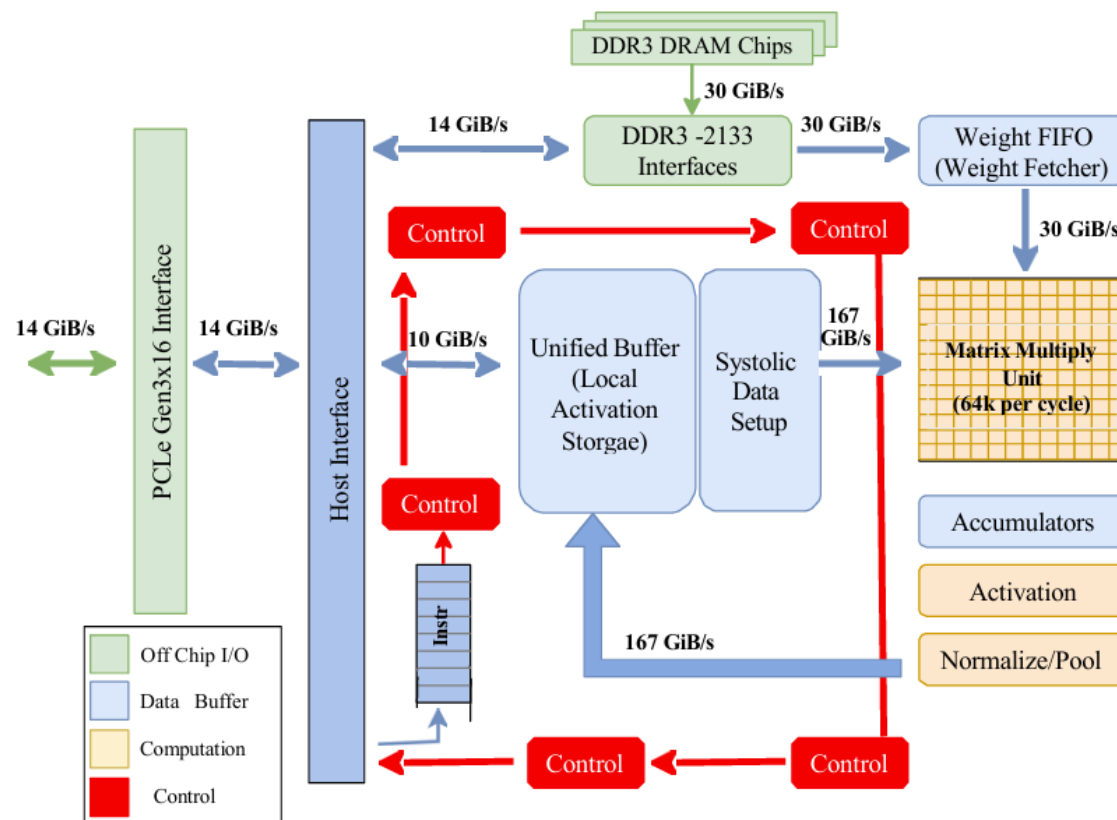


- Processing in ReRAM-based Main Memory
 - ReRAM crossbar array: enabling matrix-vector multiplication fast and efficient
 - Processing-in-Memory (PIM): reduce data movement
 - Configurable arrays: Switch ReRAM between NN accelerators and standard memory storage
 - Software/Hardware interface
- Performance and Energy efficiency
- Flexibility
- ReRAM technology hasn't been mature



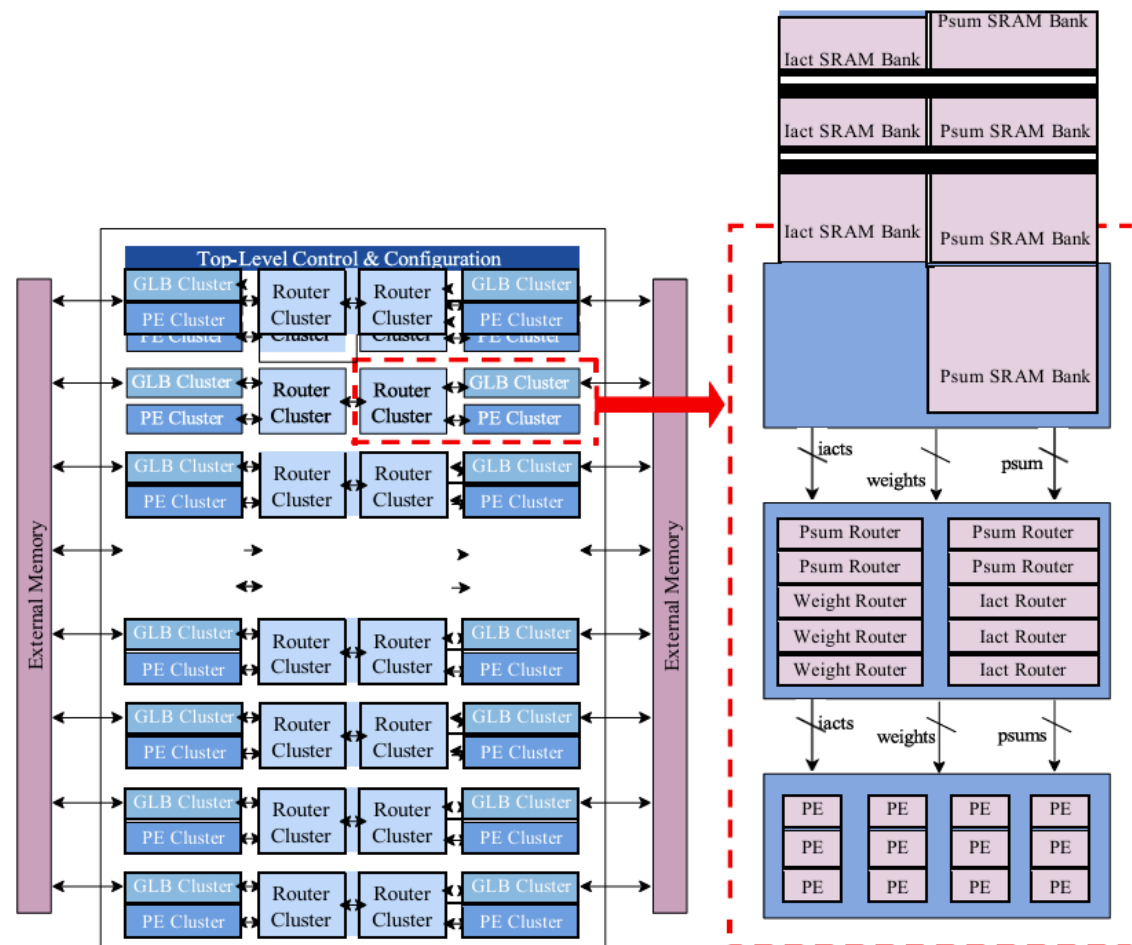
Tensor Processing Unit (TPU)

- AI ASIC developed by Google
 - Matrix Multiplication Unit: High throughput 8-bit multiply-and-add
 - Memory and Data Management: large on-chip memory: Unified Buffer for immediate data; Weight FIFO for staging weights
 - Execution model: deterministic execution model
 - Systolic Data Flow: enhancing energy efficiency and reducing read/write operations from unified buffer
 - Software compatibility: Tensorflow, JAX, PyTorch
 - Host integration via PCI
- Performance and Energy efficiency
- Less flexible
- Availability

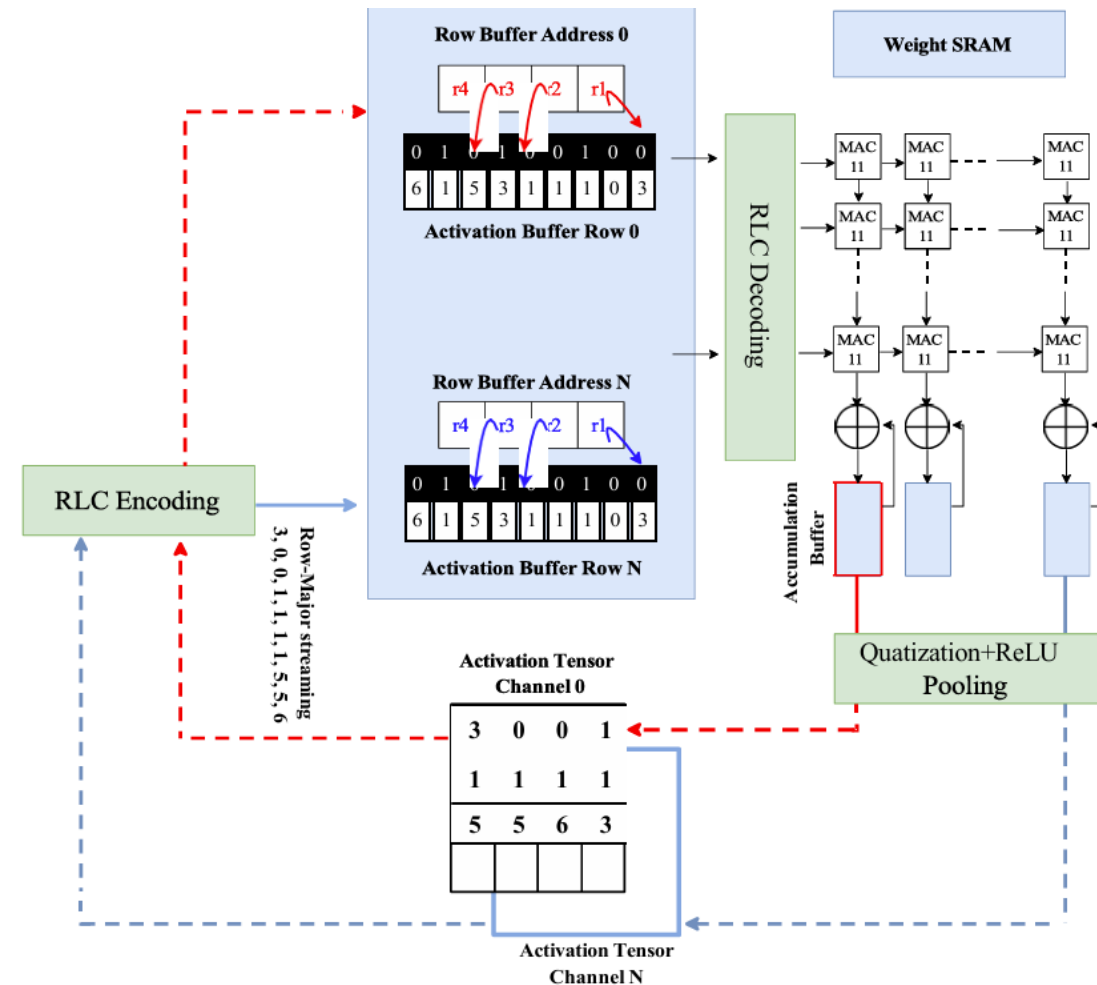


Eyeriss v2 Architecture

- AI ASIC tailored for efficient processing of compact and sparse DNN
 - Hierarchical Mesh Network (HM-NoC): handle data flow across the processor, enhancing throughput and energy efficiency
 - Processing Element (PE): tailored for sparse DNN.
 - Row-Stationary Dataflow: Optimizing data reuse and minimizing data movement.
 - Run-Length Coding: a compression technique for optimizing data storage and transfers in sparse DNN
- Suitable for mobile devices
- Limited applicability to other types of DNN



- An ASIC breakthrough in CNN
 - Systolic Array-based design
 - Run-Length Coding (RLC)
 - Adaptation for CNN workload.
 - Sparse and Lossy RLC
 - Look-ahead Snoozing technique: decrease leakage energy in activation buffers
- Suitable for mobile devices
- Limited applicability to other types of DNN



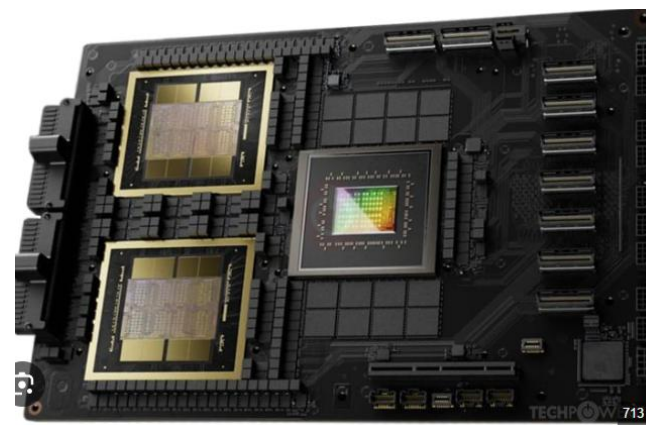
Case studies



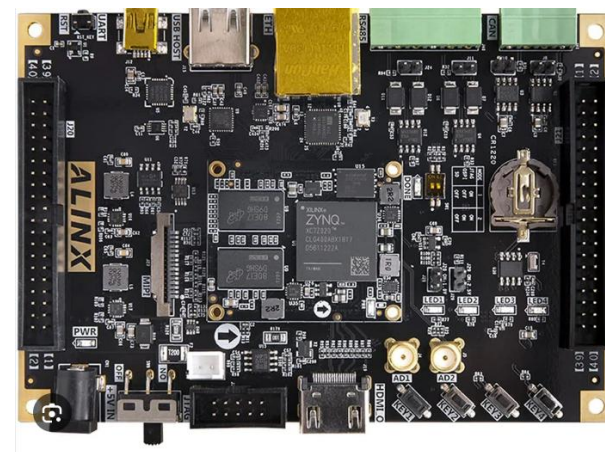
Google TPU v8



Intel Loihi 2



NVIDIA B200



ALINX AX7Z020 FPGA

A graphic on the left side of the slide. It features a dark blue background with a large, stylized circular pattern made of red dots of varying sizes, creating a halftone or dot-matrix effect. The word "HUST" is centered within this pattern in a white, bold, sans-serif font.

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THANK YOU !